



DC – AI – Routing – Switching – Security – Wireless – Automation

EANTC BIER InterOp Testing on PTX10002-36QDD

An HPE Juniper Networking Techpost

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HPE



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1 EANTC BIER InterOp Testing on PTX10002-36QDD

Suneesh Babu - 05/13/2024

BIER Interoperability testing verified between PTX10002-36QDD and other vendors during the EANTC 2024.

1.1 Introduction

BIER – Bit Index Explicit Replication provides a multicast architecture with no per-tree multicast states in the core, by having the information of the destination multicast routers within the packet. The architecture details of BIER are captured in the RFC-8279. BIER is supported on Juniper Networks PTX10002-36QDD platform (Jeffrey Zhang posted an article recently on the support brought by Express5 Forwarding ASIC: <https://juniper.github.io/techposts/bier-mvpn-in-ptx-express-5/article>) and Interop testing is successfully completed with Nokia and Huawei as part of the EANTC Interop event in Berlin and a demonstration of the interop is performed during the MPLS World Congress-2024 in Paris. This article discusses the interop test cases we performed on PTX10002-36QDD.

In the EANTC-2024 Interop event, there were three participants: Nokia 7750 SR-1, Huawei 8000 F8, and Juniper Networks PTX10002-36QDD. In the MPLS World Congress 2024 Nokia 7750 SR-1 and Juniper Networks PTX10002-36QDD BIER interop capabilities were demonstrated.

1.2 EANTC BIER Interop Topology

Interop test topology consists of four routers, one router each from Juniper Networks and Huawei and two routers from Nokia. KeySight IxNetwork is used for generating BIER multicast traffic.

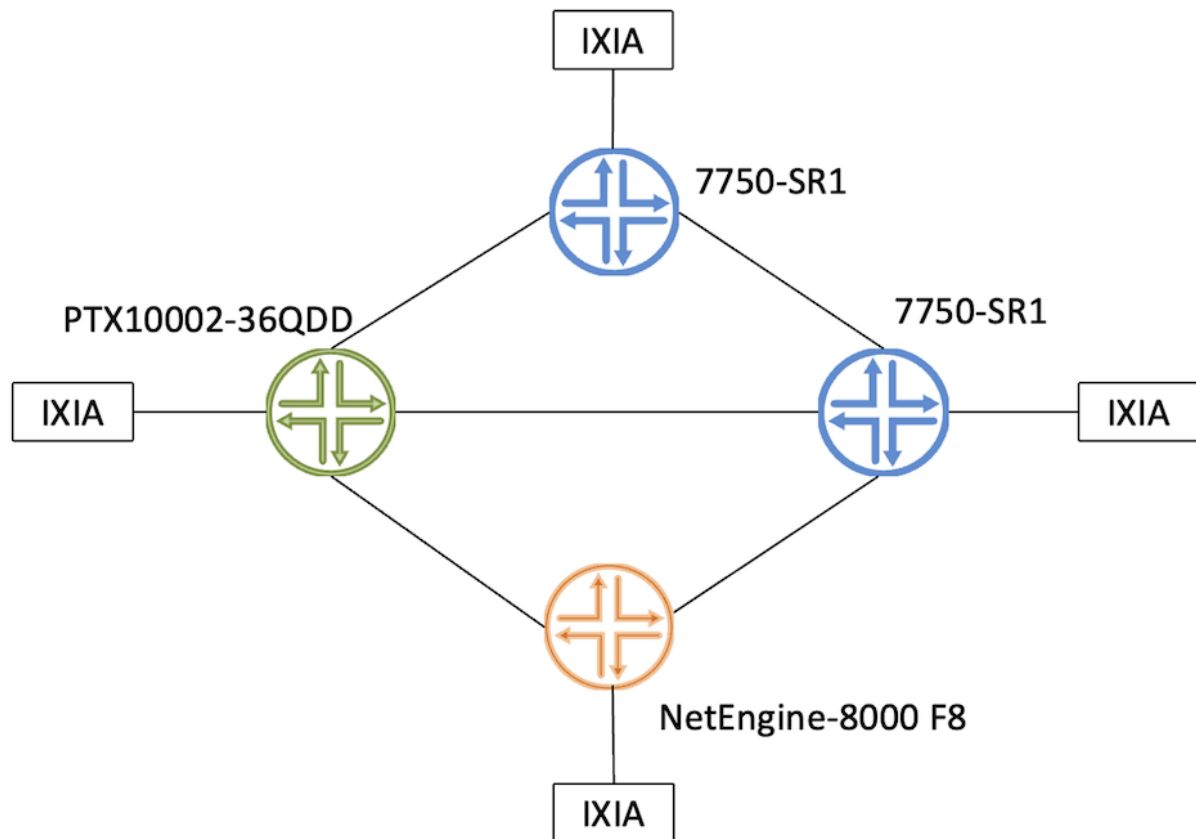


Figure 1: EANTC BIER Interop Topology

1.3 MPLS World Congress BIER Interop Demo Topology

In the MPLS World Congress BIER Demonstration, PTX10002-36QDD was peered with two Nokia 7750 devices, and BIER functionality was tested:

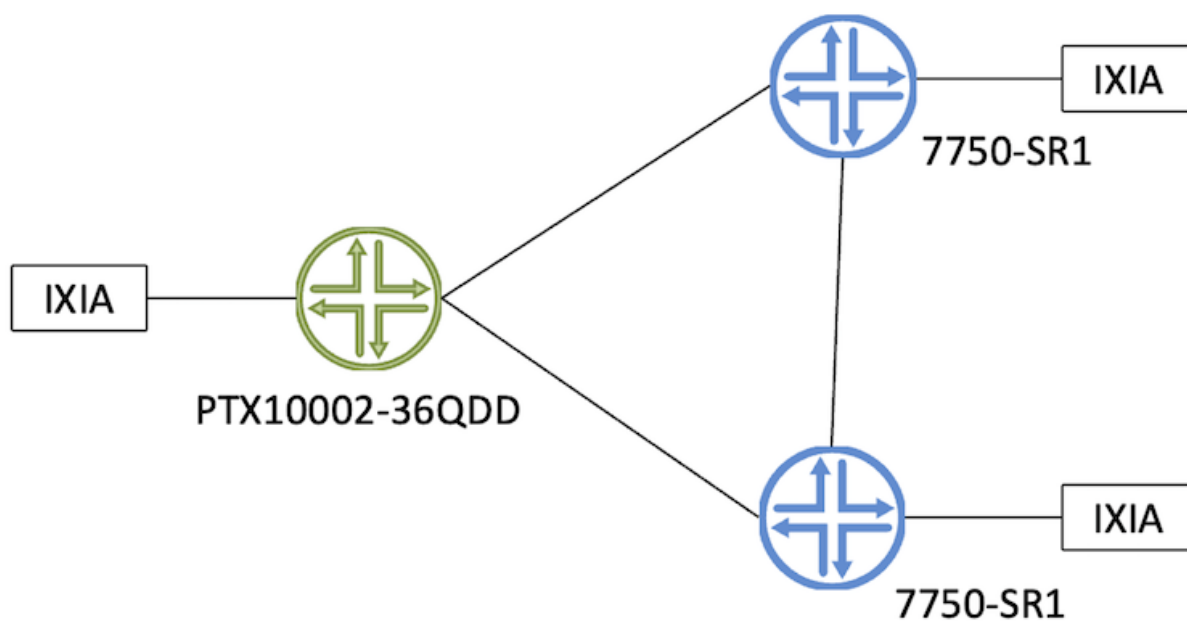


Figure 2: MWC BIER Interop Demo Topology

1.4 BIER Interop Test Cases

Interop mainly focussed on two cases, in the first test case the BIER underlay got verified with ISIS as IGP, and in the second test case, nodes' capability to replicate the multicast packet as BFR nodes were tested.

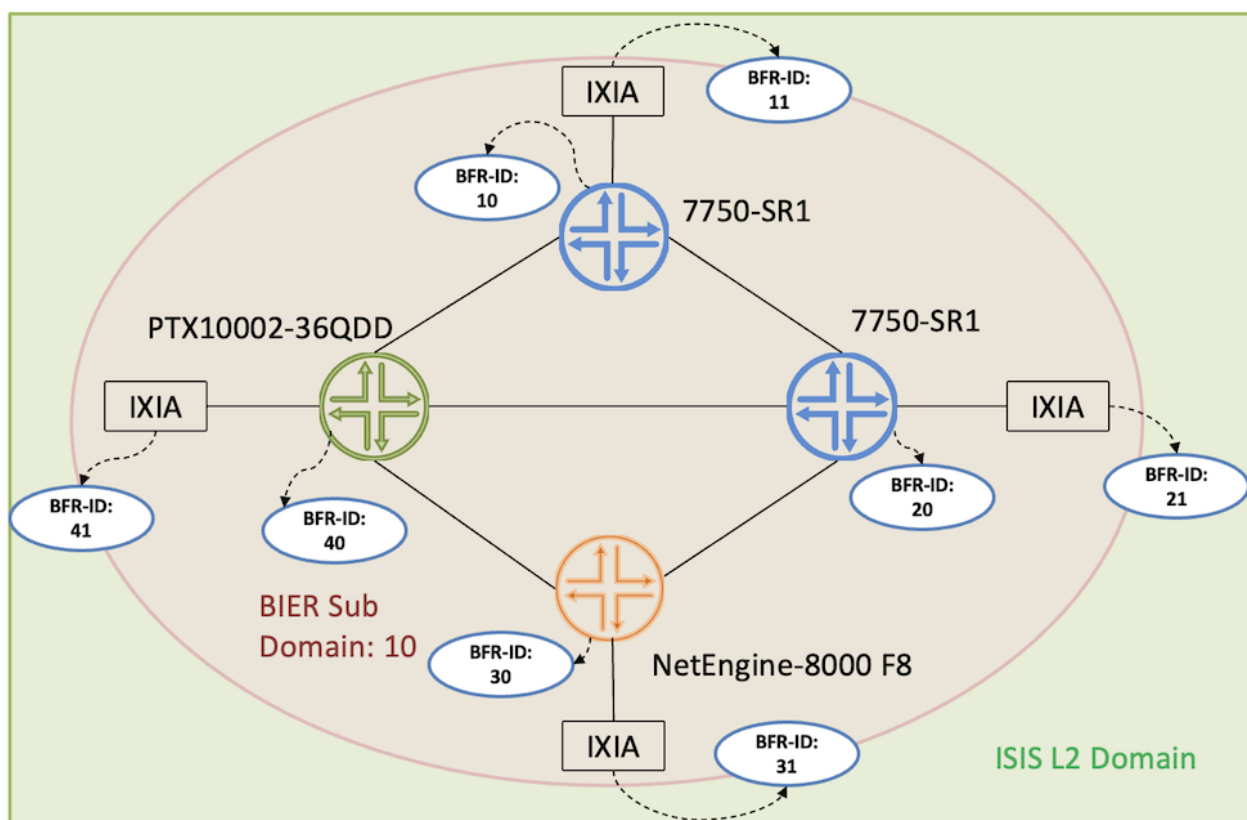


Figure 3: BIER Underlay with ISIS

1.4.1 BIER Underlay with ISIS

In the Underlay test, ISIS is used as the IGP in level-2 mode. BFR-ID got configured as shown in the below diagram with BFR-ID of 40 on PTX10002-36QDD.

BIER configuration and its association with ISIS as follows:

```

1 suneesh@JNPR-333-PTX10002-36QDD> show configuration groups BIER_LAYER
2
3 protocols {
4     bier {
5         sub-domain 10 {
6             bfr-id 40;
7             bfr-prefix 10.0.0.233;
8             encapsulation mpls bitstringlen 256 number-sets 4;
9         }
10    }
11    isis {
12        bier-sub-domain 10;
13    }
14 }
```

ISIS adjacency was formed with both Nokia devices and also with IXIA and Huawei devices as shown below:

```

1 suneesh@JNPR-333-PTX10002-36QDD> show isis adjacency |no-more
2
3 Interface System L State Hold (secs) SNPA
4 et-0/0/2.0 Nokia_58-SR1-3-SRv6 2 Up 18
5 et-0/0/20:0.41 Keys-225-333 2 Up 24
6 et-0/0/3.0 NOKIA-SR2-57 2 Up 23
7 et-0/0/8:0.0 huawei_254_NetEngine_8000_F8 2 Up 29
```

The ISIS database contains the BIER specific information including the encapsulation type, bfr-id, bit-string-length, max set identifier, and the label as highlighted below.

```

1 suneesh@JNPR-333-PTX10002-36QDD> show isis database extensive |no-more
2
3 <snip>
4 JNPR-333-PTX10002-36QDD.00-00 Sequence: 0x73, Checksum: 0xe291, Lifetime: 867 secs
5   IS neighbor: Keys-225-333.00 Metric: 10
6     Two-way fragment: Keys-225-333.00-00, Two-way first fragment: Keys-225-333.00-00
7   IS neighbor: huawei_254_NetEngine_8000_F8.00 Metric: 10
8     Two-way fragment: huawei_254_NetEngine_8000_F8.00-00, Two-way first fragment:
9       huawei_254_NetEngine_8000_F8.00-00
10   IS neighbor: NOKIA-SR2-57.00 Metric: 10
11     Two-way fragment: NOKIA-SR2-57.00-00, Two-way first fragment: NOKIA-SR2-57.00-00
12   IS neighbor: Nokia_58-SR1-3-SRv6.00 Metric: 10
13     Two-way fragment: Nokia_58-SR1-3-SRv6.00-00, Two-way first fragment: Nokia_58-SR1-3-SRv6
14       .00-00
15   IP prefix: 10.0.0.233/32 Metric: 0 Internal Up
16   IP prefix: 20.56.2.0/24 Metric: 10 Internal Up
17   IP prefix: 20.57.11.0/24 Metric: 10 Internal Up
18   IP prefix: 20.57.12.0/24 Metric: 10 Internal Up
19   IP prefix: 20.57.13.0/24 Metric: 10 Internal Up
20
21 Header: LSP ID: JNPR-333-PTX10002-36QDD.00-00, Length: 298 bytes
22   Allocated length: 1492 bytes, Router ID: 10.0.0.233
23   Remaining lifetime: 867 secs, Level: 2, Interface: 0
24   Estimated free bytes: 982, Actual free bytes: 1194
25   Aging timer expires in: 867 secs
26   Protocols: IP, IPv6
27
28 Packet: LSP ID: JNPR-333-PTX10002-36QDD.00-00, Length: 298 bytes, Lifetime : 1198 secs
29   Checksum: 0xe291, Sequence: 0x73, Attributes: 0x3 <L1 L2>
30   NLPID: 0x83, Fixed length: 27 bytes, Version: 1, Sysid length: 0 bytes
31   Packet type: 20, Packet version: 1, Max area: 0
32
33 TLVs:
34   Area address: 49.0005 (3)
35   LSP Buffer Size: 1492
36   Speaks: IP
37   Speaks: IPV6
38   IP router id: 10.0.0.233
39   IP address: 10.0.0.233
40   Hostname: JNPR-333-PTX10002-36QDD
```

```

39 Extended IS Reachability TLV, Type: 22, Length: 148
40 IS extended neighbor: Nokia_58-SR1-3-SRv6.00, Metric: default 10 SubTLV len: 26
41   IP address: 20.57.11.1
42   Neighbor's IP address: 20.57.11.2
43   Local interface index: 1027, Remote interface index: 11
44   Link MSD Advertisement Sub-TLV:Type: 15, Length: 2
45     Base MPLS Imposition MSD:Type: 1, Value: 0
46 IS extended neighbor: huawei_254_NetEngine_8000_F8.00, Metric: default 10 SubTLV len: 26
47   IP address: 20.57.13.1
48   Neighbor's IP address: 20.57.13.2
49   Local interface index: 1038, Remote interface index: 101
50   Link MSD Advertisement Sub-TLV:Type: 15, Length: 2
51     Base MPLS Imposition MSD:Type: 1, Value: 0
52 IS extended neighbor: Keys-225-333.00, Metric: default 10 SubTLV len: 26
53   IP address: 20.56.2.1
54   Neighbor's IP address: 20.56.2.2
55   Local interface index: 1059, Remote interface index: 1
56   Link MSD Advertisement Sub-TLV:Type: 15, Length: 2
57     Base MPLS Imposition MSD:Type: 1, Value: 0
58 IS extended neighbor: NOKIA-SR2-57.00, Metric: default 10 SubTLV len: 26
59   IP address: 20.57.12.1
60   Neighbor's IP address: 20.57.12.2
61   Local interface index: 1028, Remote interface index: 26
62   Link MSD Advertisement Sub-TLV:Type: 15, Length: 2
63     Base MPLS Imposition MSD:Type: 1, Value: 0
64   IP extended prefix: 20.57.11.0/24 metric 10 up
65   IP extended prefix: 20.57.13.0/24 metric 10 up
66   IP extended prefix: 10.0.0.233/32 metric 0 up
67   13 bytes of subtlvs
68   Bier-Sub-Domain: 10, Bfr-Id: 40, Bier-Algo: 0, Algo: 0
69   Max-SI: 4, BS Len: 256, Label: 17
70   IP extended prefix: 20.56.2.0/24 metric 10 up
71   IP extended prefix: 20.57.12.0/24 metric 10 up
72   Router Capability: Router ID 10.0.0.233, Flags: 0x00
73   Node MSD Advertisement Sub-TLV:Type: 23, Length: 4
74     Base MPLS Imposition MSD:Type: 1, Value: 0
75     Entropy Readable Label Depth MSD:Type: 2, Value: 8
76 No queued transmissions
77
78
79 Nokia_58-SR1-3-SRv6.00-00 Sequence: 0xad, Checksum: 0xe4bd, Lifetime: 663 secs
80 IPV4 Index: 1958
81 Node Segment Blocks Advertised:
82   Start Index : 0, Size : 2000, Label-Range: [ 20000, 21999 ]
83 IS neighbor: Keys-224-58.00 Metric: 10
84   Two-way fragment: Keys-224-58.00-00, Two-way first fragment: Keys-224-58.00-00
85   P2P IPv4 Adj-SID: 524282, Weight: 0, Flags: -BVL--
86 IS neighbor: NOKIA-SR2-57.00 Metric: 10
87   Two-way fragment: NOKIA-SR2-57.00-00, Two-way first fragment: NOKIA-SR2-57.00-00
88   P2P IPv4 Adj-SID: 524284, Weight: 0, Flags: -BVL--
89 IS neighbor: JNPR-333-PTX10002-36QDD.00 Metric: 10
90   Two-way fragment: JNPR-333-PTX10002-36QDD.00-00, Two-way first fragment: JNPR-333-PTX10002
91   -36QDD.00-00
92   P2P IPv4 Adj-SID: 524283, Weight: 0, Flags: -BVL--
93 IP prefix: 10.0.0.58/32 Metric: 0 Internal Up
94 IP prefix: 20.55.2.0/24 Metric: 10 Internal Up
95 IP prefix: 20.57.10.0/24 Metric: 10 Internal Up
96 IP prefix: 20.57.11.0/24 Metric: 10 Internal Up
97 V6 prefix: 2002::58/128 Metric: 0 Internal Up
98
99 Header: LSP ID: Nokia_58-SR1-3-SRv6.00-00, Length: 422 bytes
100   Allocated length: 422 bytes, Router ID: 10.0.0.58
101   Remaining lifetime: 663 secs, Level: 2, Interface: 1027
102   Estimated free bytes: 257, Actual free bytes: 0
103   Aging timer expires in: 663 secs
104   Protocols: IP, IPv6
105
106 Packet: LSP ID: Nokia_58-SR1-3-SRv6.00-00, Length: 422 bytes, Lifetime : 1198 secs
107   Checksum: 0xe4bd, Sequence: 0xad, Attributes: 0x3 <L1 L2>
108   NLPID: 0x83, Fixed length: 27 bytes, Version: 1, Sysid length: 0 bytes
109   Packet type: 20, Packet version: 1, Max area: 3
110
111 TLVs:
112   Area address: 49.0001.0000.0057.00 (8)
113   Speaks: IP
114   Speaks: IPV6
115   Hostname: Nokia_58-SR1-3-SRv6
116   IP router id: 10.0.0.58
117   IPv6 TE Router ID: 2002::58
118   Router Capability: Router ID 10.0.0.58, Flags: 0x00
119   SPRING Capability - Flags: 0xc0(I:1,V:1), Range: 2000, SID-Label: 20000

```

```

119 SPRING Algorithm - Algo: 0
120 Node MSD Advertisement Sub-TLV:Type: 23, Length: 4
121   Base MPLS Imposition MSD:Type: 1, Value: 12
122   Entropy Readable Label Depth MSD:Type: 2, Value: 15
123 IS neighbor: NOKIA-SR2-57.00, Internal, Metric: default 10
124 IS neighbor: JNPR-333-PTX10002-36QDD.00, Internal, Metric: default 10
125 IS neighbor: Keys-224-58.00, Internal, Metric: default 10
126 IP prefix: 10.0.0.58/32, Internal, Metric: default 0, Up
127 IP prefix: 20.55.2.0/24, Internal, Metric: default 10, Up
128 IP prefix: 20.57.10.0/24, Internal, Metric: default 10, Up
129 IP prefix: 20.57.11.0/24, Internal, Metric: default 10, Up
130 IPv6 prefix: 2002::58/128 Metric 0 Up
131 IP address: 10.0.0.58
132 IP address: 20.55.2.1
133 IP address: 20.57.10.2
134 IP address: 20.57.11.2
135 IPv6 address: 2002::58
136 Extended IS Reachability TLV, Type: 22, Length: 30
137 IS extended neighbor: NOKIA-SR2-57.00, Metric: default 10 SubTLV len: 19
138   IP address: 20.57.10.2
139   Neighbor's IP address: 20.57.10.1
140   P2P IPV4 Adj-SID - Flags:0x70(F:0,B:1,V:1,L:1,S:0,P:0), Weight:0, Label: 524284
141   P2P IPV4 Adj-SID: 524284, Weight: 0, Flags: -BVL--
142 Extended IS Reachability TLV, Type: 22, Length: 30
143 IS extended neighbor: JNPR-333-PTX10002-36QDD.00, Metric: default 10 SubTLV len: 19
144   IP address: 20.57.11.2
145   Neighbor's IP address: 20.57.11.1
146   P2P IPV4 Adj-SID - Flags:0x70(F:0,B:1,V:1,L:1,S:0,P:0), Weight:0, Label: 524283
147   P2P IPV4 Adj-SID: 524283, Weight: 0, Flags: -BVL--
148 Extended IS Reachability TLV, Type: 22, Length: 30
149 IS extended neighbor: Keys-224-58.00, Metric: default 10 SubTLV len: 19
150   IP address: 20.55.2.1
151   Neighbor's IP address: 20.55.2.2
152   P2P IPV4 Adj-SID - Flags:0x70(F:0,B:1,V:1,L:1,S:0,P:0), Weight:0, Label: 524282
153   P2P IPV4 Adj-SID: 524282, Weight: 0, Flags: -BVL--
154 IP extended prefix: 10.0.0.58/32 metric 0 up
155 21 bytes of subtlvs
156 Node SID, Flags: 0x60(R:0,N:1,P:1,E:0,V:0,L:0), Algo: SPF(0), Value: 1958
157 Bier-Sub-Domain: 10, Bfr-Id: 10, Bier-Algo: 0, Algo: 0
158   Max-SI: 16, BS Len: 256, Label: 524256
159 IP extended prefix: 20.55.2.0/24 metric 10 up
160 IP extended prefix: 20.57.10.0/24 metric 10 up
161 IP extended prefix: 20.57.11.0/24 metric 10 up
162 No queued transmissions
163
164 </snip>

```

BIER Neighborhood on PTX10002-36QDD is as follows:

```

1 suneesh@JNPR-333-PTX10002-36QDD> show bier neighbor |no-more
2
3 BIER info:
4 Subdomain ID: 10 BSL: 256
5 BIER Neighbor Label Nexthop Packets(pps) Bytes
6 100.0.0.254 874496 20.57.13.2 0( 0) 0
7 FBM:0:0:0:0:0:0:0:0:00000000
8 10.0.0.58 524256 20.57.11.2 0( 0) 0
9 FBM:0:0:0:0:0:0:0:0:00000600
10 12.0.0.225 524002 20.56.2.2 1217456( 1000) 1190671968
11 FBM:0:0:0:0:0:0:0:0:00000100:0
12 10.0.0.57 524260 20.57.12.2 0( 0) 0
13 FBM:0:0:0:0:0:0:0:0:00180000

```

BIER Routing Table is verified as follows:

```

1 suneesh@JNPR-333-PTX10002-36QDD> show route table :bier-10.inet.9 |no-more
2
3 :bier-10.inet.9: 7 destinations, 7 routes (7 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 10.0.0.57/32 *[L-ISIS/14] 00:19:25, metric 10
7   > to 20.57.12.2 via et-0/0/3.0, Push 524260
8 10.0.0.58/32 *[L-ISIS/14] 19:26:41, metric 10
9   > to 20.57.11.2 via et-0/0/2.0, Push 524256
10 12.0.0.219/32 *[L-ISIS/14] 01:29:35, metric 30
11   > to 20.57.11.2 via et-0/0/2.0, Push 524256
12 12.0.0.221/32 *[L-ISIS/14] 00:06:46, metric 30
13   > to 20.57.13.2 via et-0/0/8.0.0, Push 874496
14 12.0.0.225/32 *[L-ISIS/14] 01:29:27, metric 20

```

```

15 > to 20.56.2.2 via et-0/0/20:0.41, Push 524002
16 12.0.0.227/32 *[L-ISIS/14] 00:19:25, metric 30
17 > to 20.57.12.2 via et-0/0/3.0, Push 524260
18 100.0.0.254/32 *[L-ISIS/14] 02:27:01, metric 10
19 > to 20.57.13.2 via et-0/0/8:0.0, Push 874496

```

BIER Forwarding Table is verified as follows:

```

1 suneesh@JNPR-333-PTX10002-36QDD> show route table :bier-10-0.bier.0 |no-more
2
3 :bier-10-0.bier.0: 8 destinations, 8 routes (8 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 10/16
7      *[BIER/70] 19:26:41
8      > to 20.57.11.2 via et-0/0/2.0, Push 524256
9 11/16
10     *[BIER/70] 01:29:35
11     > to 20.57.11.2 via et-0/0/2.0, Push 524256
12 20/16
13     *[BIER/70] 00:19:25
14     > to 20.57.12.2 via et-0/0/3.0, Push 524260
15 21/16
16     *[BIER/70] 00:19:25
17     > to 20.57.12.2 via et-0/0/3.0, Push 524260
18 30/16
19     *[BIER/70] 02:27:01
20     > to 20.57.13.2 via et-0/0/8:0.0, Push 874496
21 31/16
22     *[BIER/70] 01:23:42
23     > to 20.57.13.2 via et-0/0/8:0.0, Push 874496
24 40/16
25     *[BIER/70] 22:27:15
26     to table mpls.0
27 41/16
28     *[BIER/70] 01:29:27
29     > to 20.56.2.2 via et-0/0/20:0.41, Push 524002

```

MPLS Routing Table is as follows:

```

1 <snip>
2
3 mpls.0: 11 destinations, 11 routes (11 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 0 *[MPLS/0] 23:47:19, metric 1
7      to table inet.0
8 0(S=0) *[MPLS/0] 23:47:19, metric 1
9      to table mpls.0
10 1 *[MPLS/0] 23:47:19, metric 1
11      Receive
12 2 *[MPLS/0] 23:47:19, metric 1
13      to table inet6.0
14 2(S=0) *[MPLS/0] 23:47:19, metric 1
15      to table mpls.0
16 13 *[MPLS/0] 23:47:19, metric 1
17      Receive
18 17 *[BIER/70] 00:20:01
19      Multicast (IPv4) Composite
20 18 *[BIER/70] 22:27:51
21      Discard
22 19 *[BIER/70] 22:27:51
23      Discard
24 20 *[BIER/70] 22:27:51
25      Discard
26 </snip>

```

1.5 BIER Multicast Replication

In the BIER multicast replication test, IXIA acted as the BFIR and BFER roles, with routers acted as transit or BFR role. IXIA send multicast traffic with BIER header having receivers on the IXIA nodes. As shown below, once the multicast traffic with BIER header having destination set with BFR-ID of 11, 21 and 31 hits

the PTX10002-36QDD router, it looks into the respective BIER forwarding table and replicates the multicast packets to three different destinations.

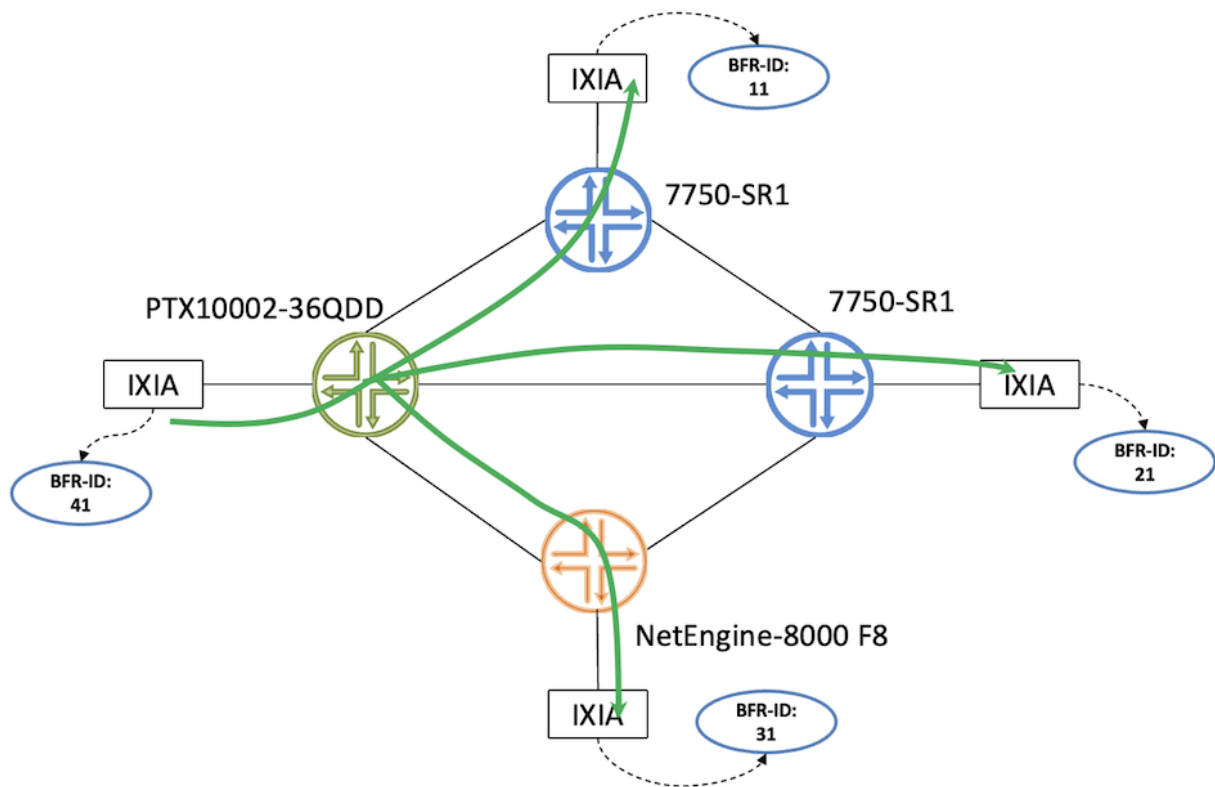


Figure 4: BIER Multicast Replication

As shown below the packets got replicated to three neighbors

```

1 suneesh@JNPR-333-PTX10002-36QDD> show interfaces extensive et-0/0/2 |match pps
2   Input packets: 1312 0 pps
3   Output packets: 985697 1001 pps
4   Input packets: 1 1 pps
5   Output packets: 978467 1000 pps
6
7 suneesh@JNPR-333-PTX10002-36QDD> show interfaces extensive et-0/0/3 |match pps
8   Input packets: 1314198 0 pps
9   Output packets: 542453 1002 pps
10  Input packets: 2626212 0 pps
11  Output packets: 538301 1000 pps
12
13 suneesh@JNPR-333-PTX10002-36QDD> show interfaces extensive et-0/0/8:0 |match pps
14   Input packets: 559926 0 pps
15   Output packets: 1009698 1001 pps
16   Input packets: 1116750 0 pps
17   Output packets: 1002464 1000 pps

```

The test also got repeated with replication happening at Nokia and Huawei nodes.

1.6 Conclusion

PTX10002-36QDD demonstrated its BIER interop capability on multicast replication with both Nokia and Huawei and the same is verified by EANTC and demonstrated in MWC-2024.

1.7 Useful Links

- <https://eantc.de/wp-content/uploads/2023/12/EANTC-MPLSSDNInterop2024-TestReport-v1.3.pdf>
- <https://datatracker.ietf.org/wg/bier/documents/>
- <https://juniper.github.io/techposts/bier-mvpn-in-ptx-express-5/article>
- <https://juniper.github.io/techposts/bier-for-sr-multicast-cheers/article>
- <https://juniper.github.io/techposts/introducing-ptx10002-36qdd/article>

1.8 Glossary

- BIER Bit Index Explicit Replication
- BFR Bit Forwarding Router
- BFER Bit Forwarding Egress Router
- BFIR Bit Forwarding Ingress Router
- BIFT Bit Index Forwarding Table
- BIRT Bit Index Forwarding Table
- BSL Bit String Length
- BFR-ID BFR Identifier
- EANTC European Advanced Networking Test Center
- FBM Forwarding Bit Mask
- MWC MPLS World Congress
- RFC Request For Comments

1.9 Acknowledgements

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